Build one spatial ANOVA map panel

Description

Builds a single `ggplot2` map panel showing the Shapley-adjusted percentage variance explained by Associations for one combination of continent, spatial scale, and taxonomic resolution. Grid rectangles are coloured by association percentage (viridis gradient, grey when no result). Fossil core locations are overlaid as points when available.

Usage

```
plot_spatial_anova_map_panel(
  sel_continent,
  sel_scale,
  sel_resolution,
  xlim_val,
  ylim_val,
  data_grid_with_results,
  data_coords_per_unit,
  sf_world,
  graphical_options,
  show_x_axis = TRUE,
  show_y_axis = TRUE,
  col_label = NULL,
  row_label = NULL
)
```

Arguments

`sel_continent`	Character scalar. Continent identifier (e.g. `"europe"`, `"america"`, `"asia"`).
`sel_scale`	Character scalar. Spatial scale identifier (e.g. `"continental"`, `"regional"`, `"local"`).

`sel_resolution`	Character scalar. Taxonomic resolution identifier (e.g. `"genus"`, `"family"`, `"functional_type"`).
`xlim_val`	Numeric vector of length 2. Longitude limits for `ggplot2::coord_sf()` (western, eastern).
`ylim_val`	Numeric vector of length 2. Latitude limits for `ggplot2::coord_sf()` (southern, northern).
`data_grid_with_results`	Data frame. Spatial grid joined with ANOVA association percentages; must contain columns `continent`, `scale`, `taxonomic_resolution`, `x_min`, `x_max`, `y_min`, `y_max`, `assoc_pct`.
`data_coords_per_unit`	Data frame. Per-unit fossil core coordinates; must contain columns `continent`, `scale`, `coords` (list-column of data frames with `coord_long` and `coord_lat`).
`sf_world`	`sf` object. World polygon basemap used as the background layer.
`graphical_options`	Named list of canvas settings (e.g. from `load_active_config_value("graphical")`); must contain `width`, `height`, `units`, `dpi`, `bg`.
`show_x_axis`	Logical scalar. When `FALSE`, x-axis text and tick marks are hidden. Default `TRUE`.
`show_y_axis`	Logical scalar. When `FALSE`, y-axis text and tick marks are hidden. Default `TRUE`.
`col_label`	Character scalar or `NULL`. When not `NULL`, added as a bold column-header title above the panel via `ggplot2::ggtitle()`. Default `NULL`.
`row_label`	Character scalar or `NULL`. When not `NULL`, added as a bold rotated y-axis title for the panel row. Default `NULL`.

Details

The function applies `ggview::canvas()` using the supplied `graphical_options` and ends with `ggplot2::coord_sf()` so that the continent-specific bounding box takes precedence over the automatic bounding box added by `ggplot2::geom_sf()`.

Value

A `ggplot` object representing one map panel.

See Also

`extract_jsdm_variance_fractions`, `compute_shapley_variance_components`